Auto Ticket Classification Using Flow Designer

Project Documentation

1. Introduction

Project Title:
Auto Ticket Classification Using Flow Designer

Team Members:

1. **Team Member 1** – Flow Designer Configuration & Development
2. **Team Member 2** – Testing, Validation & Documentation

2. Project Overview

Purpose

The purpose of this project is to automate the classification of IT support tickets in a school environment. The system uses ServiceNow Flow Designer to analyze the **Short Description** and **Description** of newly created incidents and automatically identify the appropriate category and subcategory. This reduces manual work and helps the IT support team process tickets faster and more consistently.

Features

- Automatic incident classification
- Analysis of Short Description and Description
- Keyword-based classification
- Automatic category assignment
- Automatic subcategory assignment
- Flow Designer automation
- Support for common issues such as:
    - Wi-Fi/Network problems
    - Projector problems
    - Password issues
    - Slow computer problems
- Automatic incident record update
- Easy modification and maintenance of classification rules

3. Architecture

System Architecture

The system follows a simple automation-based architecture:

Student/Teacher → ServiceNow Incident → Flow Designer → Keyword & Condition Analysis → Classification → Category/Subcategory Update → IT Support Team

Input

The user creates an IT incident containing:

- Short Description
- Description
- Caller/User details
- Other incident information

Processing

When a new incident is created, **Flow Designer** is triggered. It reads the incident description and checks predefined keywords and conditions.

Output

The system automatically updates the incident with the appropriate:

- Category
- Subcategory

The classified ticket is then available for the IT support team.

Database

The incident information is stored in the **ServiceNow Incident table**. The Flow Designer reads and updates the incident record during the classification process.

4. Setup Instructions

Prerequisites

- ServiceNow account
- ServiceNow development instance
- Access to Flow Designer
- Internet connection
- Basic knowledge of ServiceNow Incident Management

Installation / Configuration

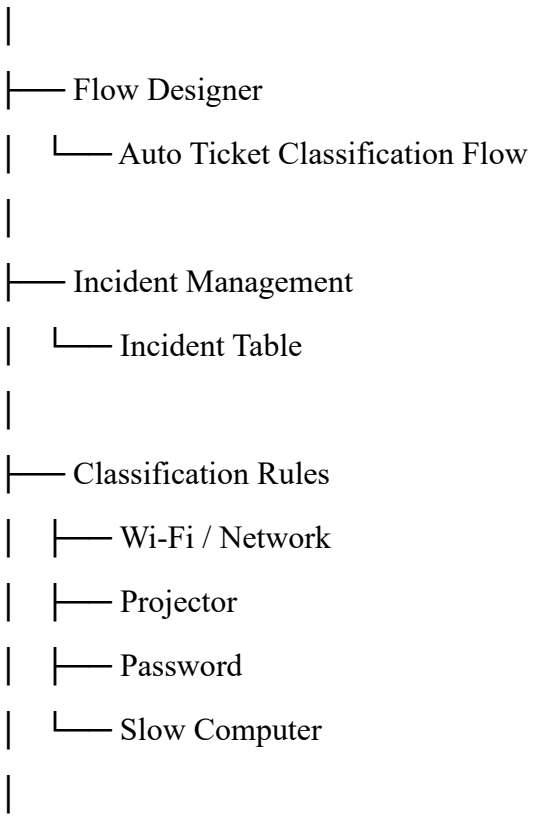
1. Log in to the ServiceNow development instance.
2. Open the **Incident** module.
3. Open **Flow Designer**.
4. Create a new flow.
5. Configure the trigger as **Incident Created**.
6. Add conditions for different IT issues.
7. Configure keywords for each issue type.
8. Add actions to update the Category and Subcategory.
9. Save and activate the flow.
10. Create test incidents and verify the classification.

5. Folder Structure

Since the project is developed inside ServiceNow, it does not use a traditional client/server folder structure.

ServiceNow Project Structure

Auto Ticket Classification



├── Conditions

│ └── Keyword-based Conditions

│

└── Testing

└── Test Incidents

Flow Designer

Contains the main automation flow, trigger, conditions, and actions.

Incident Table

Stores the incoming IT support incidents and their classification results.

Classification Rules

Contains predefined keywords and conditions used to identify different issue types.

6. Running the Application

The project does not require commands such as npm start because it runs within the ServiceNow platform.

Steps to Run

1. Open the ServiceNow development instance.
2. Log in with an authorized account.
3. Navigate to **Flow Designer**.
4. Verify that the Auto Ticket Classification flow is active.
5. Navigate to the Incident module.
6. Create a new incident.
7. Enter an issue in the Short Description and Description fields.
8. Submit the incident.
9. The Flow Designer automatically processes the incident.
10. Check the Category and Subcategory fields.

Example

Short Description:
Wi-Fi is not connecting

Result:
Category → **Network**
Subcategory → **Wi-Fi**

7. API Documentation

The project does not use external APIs.

The classification process is handled internally by **ServiceNow Flow Designer**.

Component	Method/Action	Purpose
Incident	Create	Creates a new IT incident
Flow Designer	Trigger	Starts the classification flow
Incident Record	Read	Reads Short Description and Description
Classification Rules	Condition Check	Identifies the issue type
Incident Record	Update	Updates Category and Subcategory

External API: Not Applicable

8. Authentication

Authentication is handled by the **ServiceNow platform**.

Users must log in using authorized ServiceNow accounts to access the incident management system.

Security Features

- User authentication
- Role-based access
- Authorized access to incidents
- Controlled access to Flow Designer
- ServiceNow platform security

Only authorized users can create, view, or modify incidents according to their assigned permissions.

9. User Interface

The project uses the ServiceNow user interface for incident management.

Main UI Screens

1.Incident Creation Screen
Used by students or teachers/support users to enter the issue details.

2. Incident List
Displays submitted incidents and their current status.

3. Incident Details
Shows the Short Description, Description, Category, and Subcategory.

4. Flow Designer Screen
Displays the configured automation flow, triggers, conditions, and actions.

5. Classification Result
Shows the automatically assigned category and subcategory.

Example

Short Description: Projector is not working

Description: The classroom projector is not displaying the screen.

Category: Hardware

Subcategory: Projector

10. Testing

Testing was performed to verify that the Flow Designer correctly classifies different types of IT incidents.

Test Cases

Test Case	Input Issue	Expected Result	Status
TC01	Wi-Fi not working	Network / Wi-Fi	Pass
TC02	Projector not working	Hardware / Projector	Pass
TC03	Forgot password	Account / Password	Pass
TC04	Computer is slow	Computer / Performance	Pass
TC05	Description-based issue	Correct classification	Pass

Test Case	Input Issue	Expected Result	Status
TC06	Multiple incidents	Correct classification for each	Pass
TC07	Incident record update	Category/Subcategory updated	Pass
TC08	Flow trigger	Flow starts on incident creation	Pass

Testing Result

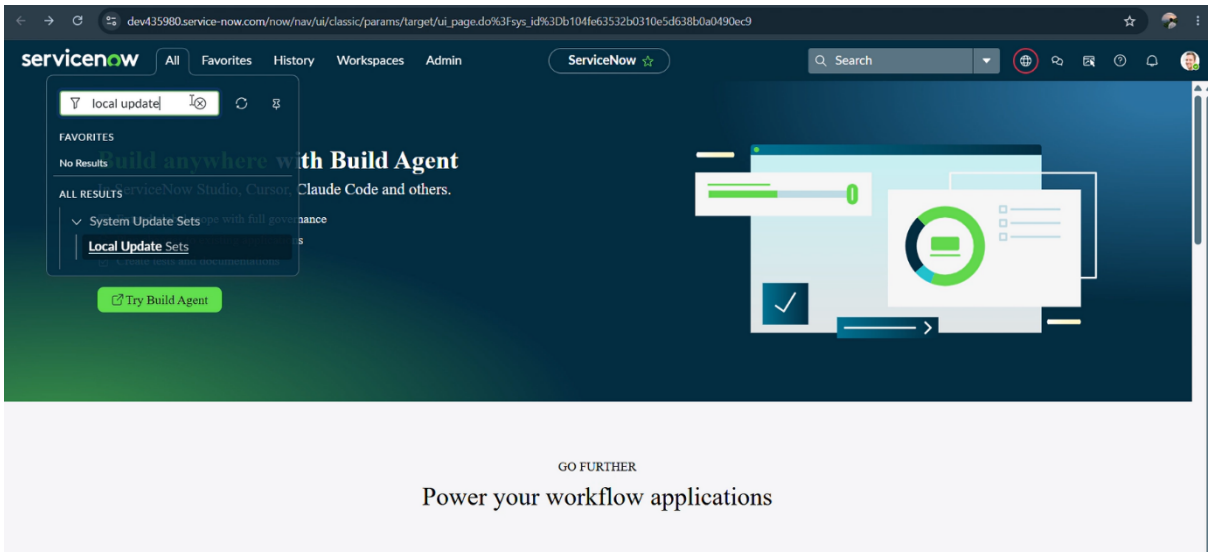
All major test scenarios were successfully executed. The Flow Designer correctly processed the tested incidents and updated the classification fields.

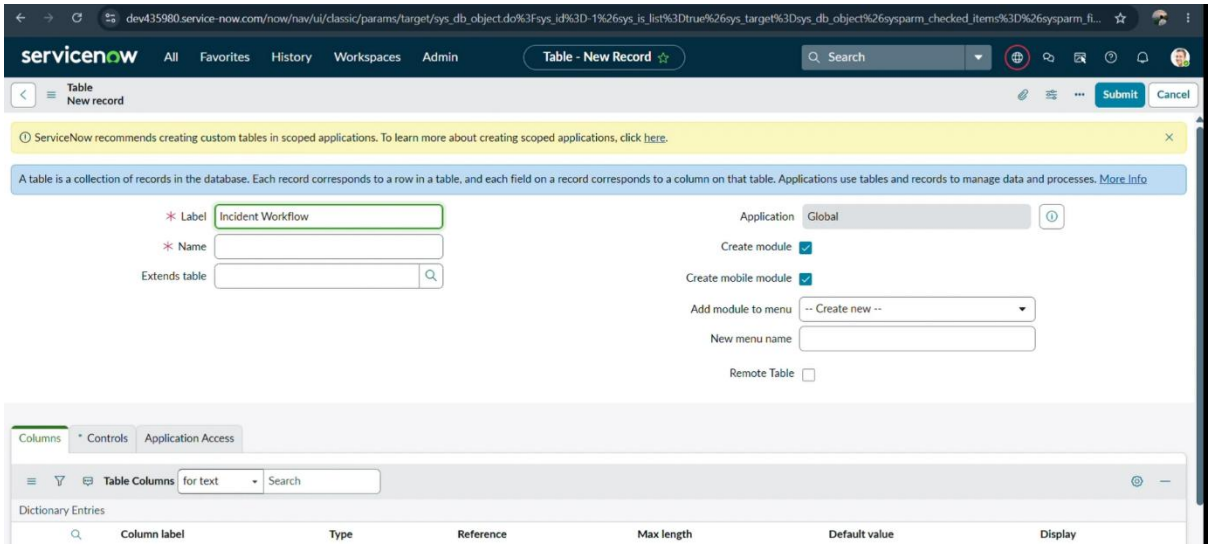
Overall Test Result: PASS

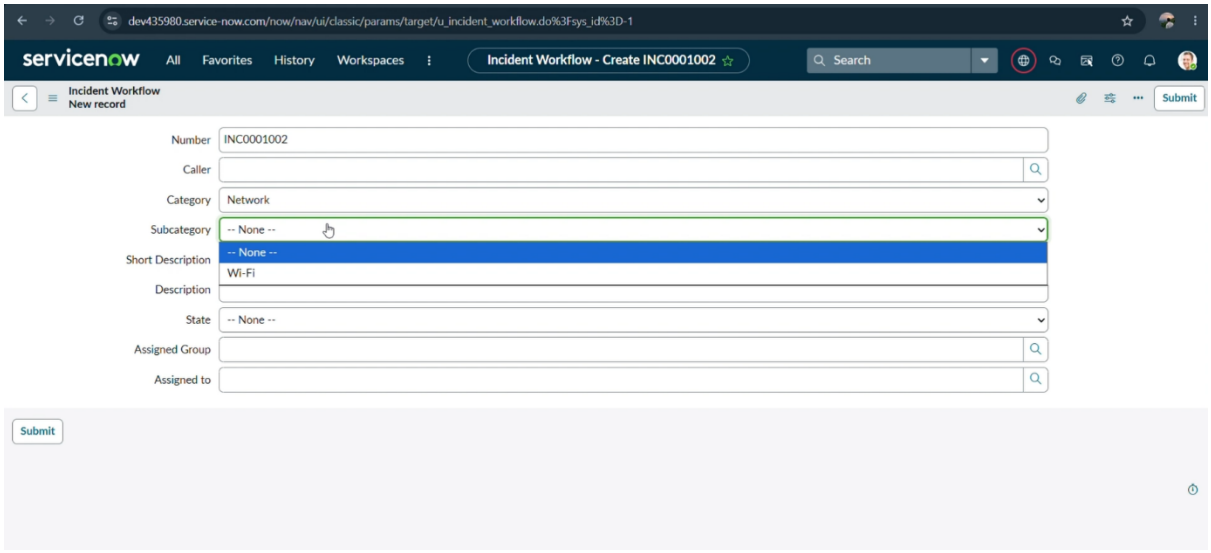
11. Screenshots or Demo

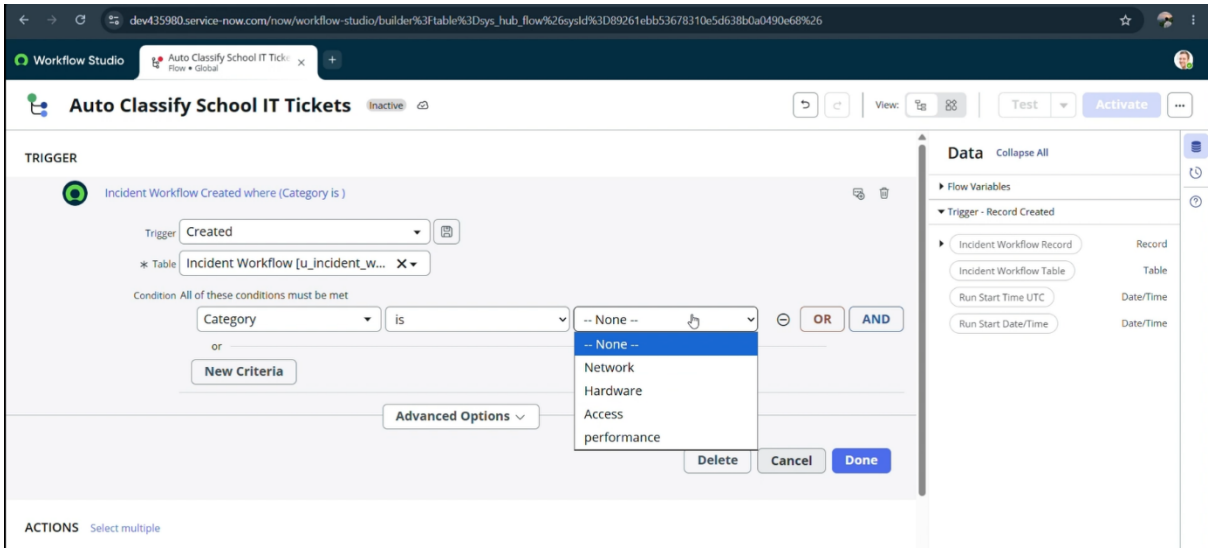
The following screenshots can be included in the project documentation:

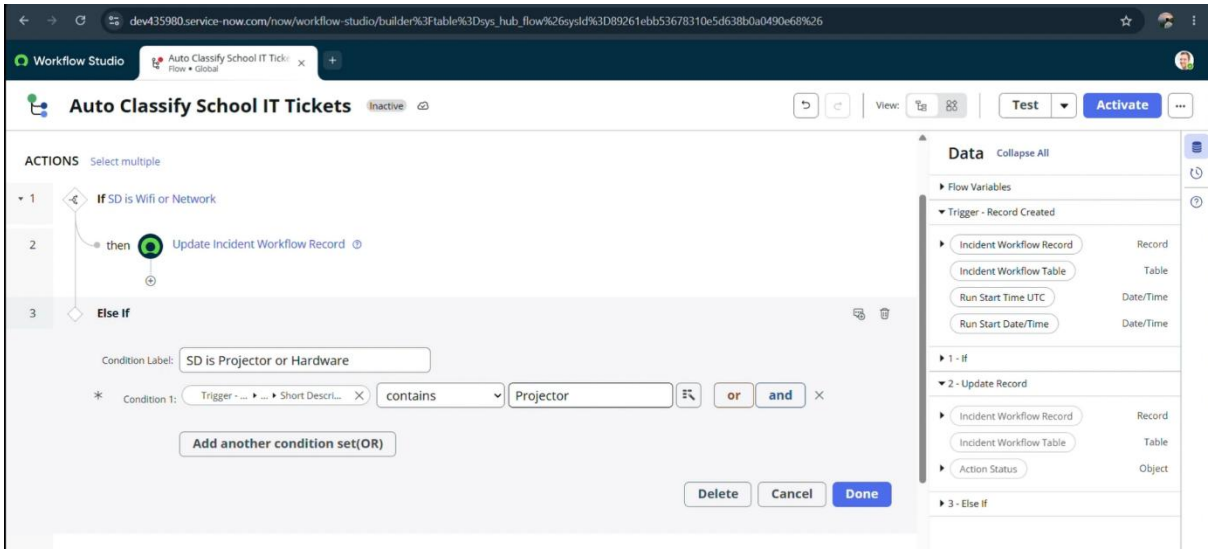
ServiceNow Login Screen

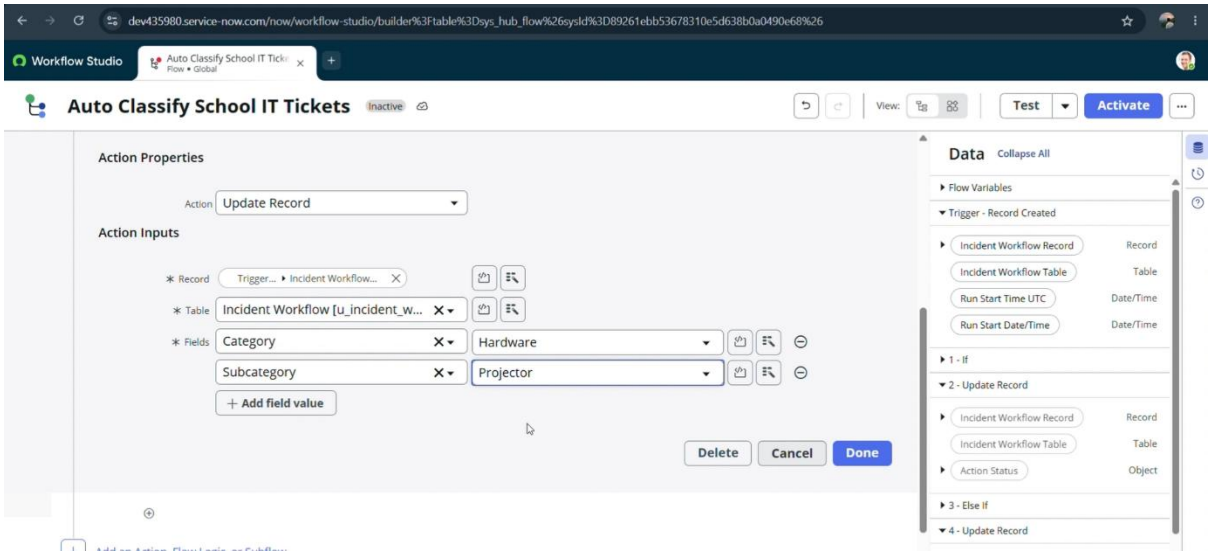


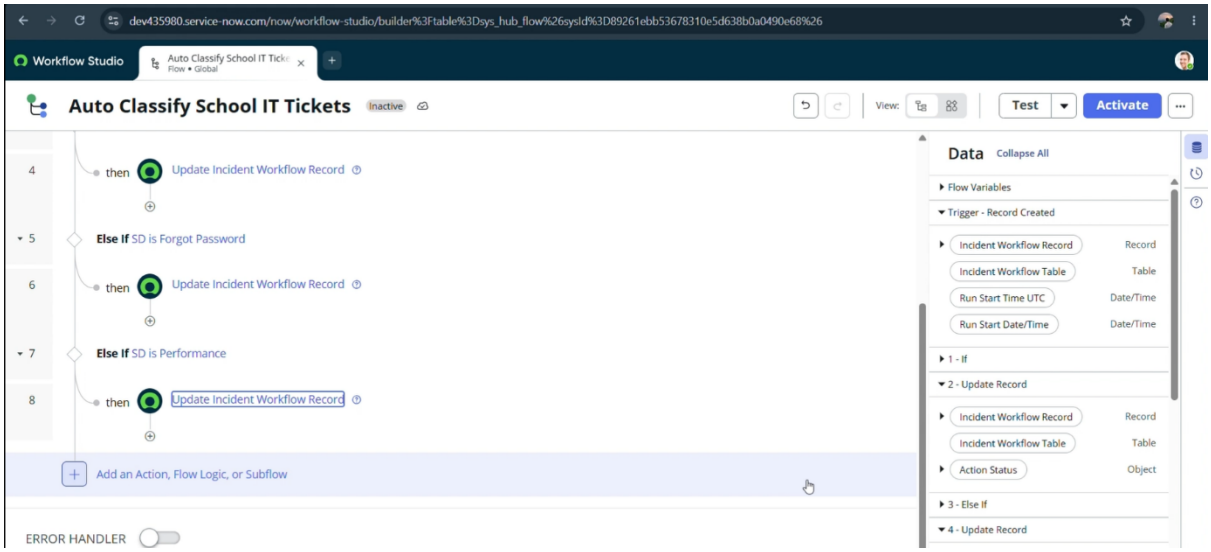

servicenow
All
Favorites
History
Workspaces
Admin
Table - New Record
Search
Table
New record
Submit
Cancel
ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click here.
A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. More Info
Label
Incident Workflow
Name
Extends table
Application
Global
Create module
Create mobile module
Add module to menu
-- Create new --
New menu name
Remote Table
Columns
Controls
Application Access
Table Columns
for text
Search
Dictionary Entries
Column label
Type
Reference
Max length
Default value
Display


servicenow
All
Favorites
History
Workspaces
Incident Workflow - Create INC0001002
Search
Incident Workflow
New record
Submit
Number
INC0001002
Caller
Category
Network
Subcategory
-- None --
-- None --
Wi-Fi
Short Description
Description
State
-- None --
Assigned Group
Assigned to
Submit

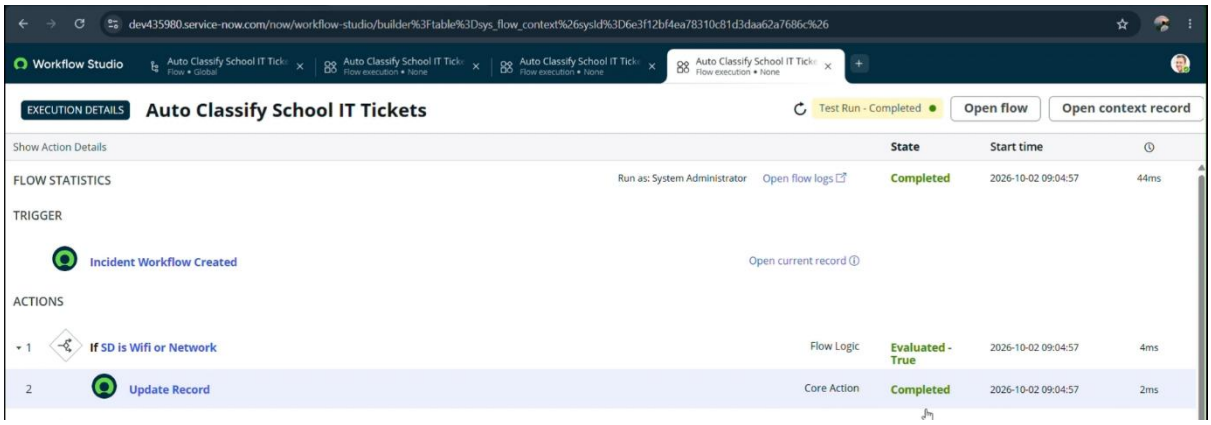

Workflow Studio
Auto Classify School IT Tickets
Inactive
View:
Test
Activate
TRIGGER
Incident Workflow Created where (Category is)
Trigger
Created
Table
Incident Workflow [u_incident_w...
Condition
All of these conditions must be met
Category
is
-- None --
OR
AND
or
New Criteria
Advanced Options
-- None --
Network
Hardware
Access
performance
Delete
Cancel
Done
ACTIONS
Select multiple
Data
Collapse All
Flow Variables
Trigger - Record Created
Incident Workflow Record
Record
Incident Workflow Table
Table
Run Start Time UTC
Date/Time
Run Start Date/Time
Date/Time

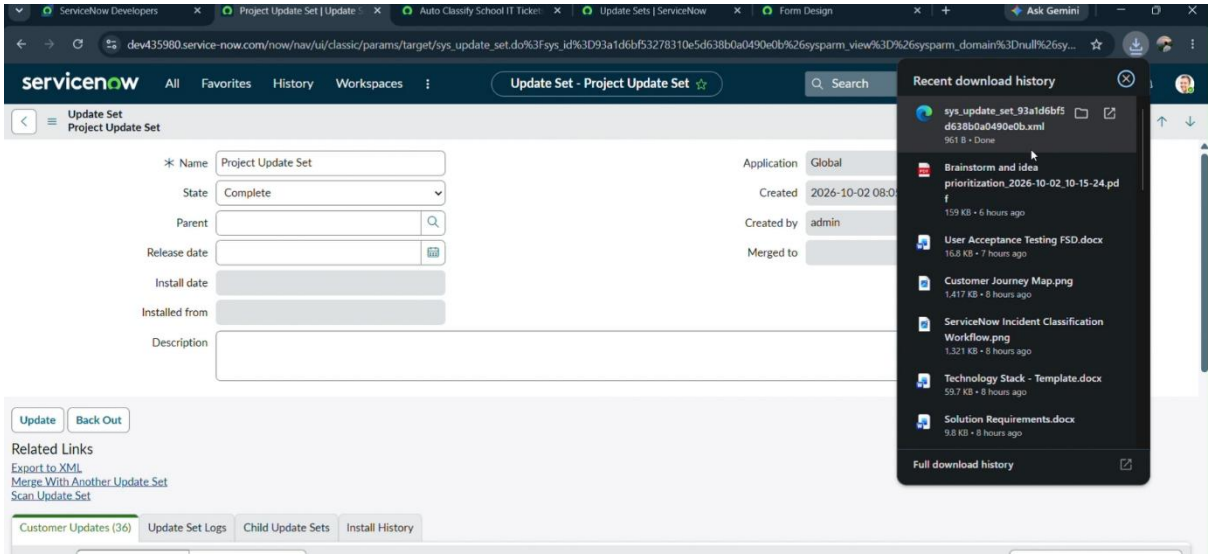

Workflow Studio
Auto Classify School IT Tickets
Inactive
Test
Activate
ACTIONS
Select multiple
1 If SD is Wifi or Network
2 then Update Incident Workflow Record
3 Else If
Condition Label: SD is Projector or Hardware
Condition 1: Trigger - ... Short Descri... contains Projector or and
Add another condition set(OR)
Delete
Cancel
Done
Data
Collapse All
Flow Variables
Trigger - Record Created
Incident Workflow Record
Incident Workflow Table
Run Start Time UTC
Run Start Date/Time
1 - If
2 - Update Record
Action Status
3 - Else If


Workflow Studio
Auto Classify School IT Tickets
Inactive
Test
Activate
Action Properties
Action Update Record
Action Inputs
Record Trigger... Incident Workflow...
Table Incident Workflow [u_incident_w...
Fields Category Hardware
Subcategory Projector
Add field value
Delete
Cancel
Done
Data
Collapse All
4 - Update Record


Workflow Studio
Auto Classify School IT Tickets
Inactive
Test
Activate
4 then Update Incident Workflow Record
5 Else If SD is Forgot Password
6 then Update Incident Workflow Record
7 Else If SD is Performance
8 then Update Incident Workflow Record
Add an Action, Flow Logic, or Subflow
ERROR HANDLER
Data
Collapse All

Testing Result





12. Known Issues

- Classification depends on predefined keywords and conditions.
- Incorrect or incomplete descriptions may affect classification.
- New types of IT problems require additional classification rules.
- Similar keywords may sometimes require more specific conditions.
- The system depends on the availability of the ServiceNow platform.
- The project does not use machine learning, so it cannot automatically learn new classifications.

13. Future Enhancements

The project can be improved with the following features:

- Add more IT issue categories and keywords.
- Add automatic assignment to the appropriate support team.
- Add automatic priority assignment.
- Create dashboards for ticket classification statistics.
- Add notifications for newly classified tickets.
- Add more advanced text analysis.
- Add multilingual ticket classification.
- Add feedback-based improvement of classification rules.
- Integrate knowledge-base suggestions.
- Extend the system to other departments such as HR, Administration, and Facilities.

Conclusion

The **Auto Ticket Classification Using Flow Designer** project provides a simple and effective way to automate IT ticket classification. By analyzing the Short Description and Description of incidents and applying predefined conditions, the system automatically assigns appropriate categories and subcategories. This reduces repetitive manual work, improves consistency, and helps the school IT support team process incidents more efficiently.